

7(a)	gene – a length of DNA / a sequence of DNA nucleotides, that codes for a particular, protein / polypeptide ; locus – the position of a, gene / allele, on a chromosome ;	2																									
7(b)	parent genotype (AaBb × AaBb) parent phenotype brown (brown) ; gametes AB Ab aB ab × (AB Ab aB ab) ; <table><tr><td></td><td>AB</td><td>Ab</td><td>aB</td><td>ab</td></tr><tr><td>AB</td><td>AABB</td><td>AABb</td><td>AaBB</td><td>AaBb</td></tr><tr><td>Ab</td><td>AABb</td><td>AAbb</td><td>AaBb</td><td>Aabb</td></tr><tr><td>aB</td><td>AaBB</td><td>AaBb</td><td>aaBB</td><td>aaBb</td></tr><tr><td>ab</td><td>AaBb</td><td>Aabb</td><td>aaBb</td><td>aabb</td></tr></table> ;; offspring phenotypes linked to genotypes ; ratio – 9 brown : 3 black : 4 white ;		AB	Ab	aB	ab	AB	AABB	AABb	AaBB	AaBb	Ab	AABb	AAbb	AaBb	Aabb	aB	AaBB	AaBb	aaBB	aaBb	ab	AaBb	Aabb	aaBb	aabb	6
	AB	Ab	aB	ab																							
AB	AABB	AABb	AaBB	AaBb																							
Ab	AABb	AAbb	AaBb	Aabb																							
aB	AaBB	AaBb	aaBB	aaBb																							
ab	AaBb	Aabb	aaBb	aabb																							

Question	Answer	Marks
7(c)	any three from: 1 (base), substitution / deletion / insertion or frame shift ; 2 <i>ref. to</i> change in, primary / secondary / tertiary, structure (of polypeptide / protein) or change in, 3D / active site, shape ; 3 <i>ref. to</i> stop codon ; 4 (so) no / inactive, tyrosinase produced ; 5 tyrosine not converted to, DOPA / dopaquinone ; 6 melanin not formed ;	3